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Fields of Concentration:
Primary Fields: International Economics, Spatial Economics
Secondary Field: Environmental Economics

Qualifying Examinations Completed:
2022 (Oral): Macroeconomics, Spatial Economics
2021 (Written): Microeconomics, Macroeconomics

Dissertation Title: *Essays on the Spatial Economics of Environmental and Energy Policy*

Committee:
Professor Costas Arkolakis (Chair)
Professor Kenneth Gillingham
Professor Amit Khandelwal
Professor Samuel Kortum

Education:
Ph.D., Economics, Yale University, 2027 (expected)
M.Phil., Economics, Yale University, 2023
M.A., Economics, Yale University, 2023
M.A., Economics, Seoul National University, 2020
B.A., Economics, Seoul National University, 2018

Fellowships, Honors, and Awards:
Department Travel Grants (two awards), Yale University, 2026
Allan Shelden III Fellowship, Yale University, 2025
Ryoichi Sasakawa Young Leaders Fellowship (SYLFF), Yale University, 2024
Doctoral Fellowship, Cowles Foundation Fellowship, Yale University, 2020–2026
Research and Lecture Scholarship, Seoul National University, 2019–2020
Brain Korea 21 Plus Research Scholarship, Seoul National University, 2018–2020
Seoul National College of Commerce Alumni Association Scholarship, 2017
Seoul National University Foundation Scholarship, 2014
Academic Merit Scholarship, Seoul National University, 2012–2017

Research Grants:

Data Acquisition Grant, Department of Economics, Yale University, 2026 (USD 3,000)
Data Acquisition Grant, Economic Growth Center, Yale University, 2025 (USD 5,000)
SYLFF Research Award, Yale University, 2024 (USD 24,078)
SYLFF Research Award, Yale University, 2023 (USD 5,913)

Teaching Experience:

Fall 2026, Guest Lecturer, Urban and Environmental Economics (G), Yale University
Spring 2026, Teaching Assistant to Profs. Giuseppe Moscarini and Aleh Tsyvinski,
Introductory Macroeconomics (U), Yale University
Fall 2024, Guest Lecturer, Urban and Environmental Economics (G), Yale University
Fall 2024, Teaching Assistant to Prof. Ana Cecilia Fieler,
International Finance (U), Yale University
Spring 2023, Teaching Assistant to Prof. Giuseppe Moscarini,
Macroeconomics (G), Yale University
Fall 2022, Teaching Assistant to Profs. Giuseppe Moscarini and Fabrizio Zilibotti,
Introductory Macroeconomics (U), Yale University
Spring 2020, Teaching Assistant to Prof. Soyoung Kim,
Recent Trends in Macroeconomics (G), Seoul National University
Fall 2019, Teaching Assistant to Prof. Woong Yong Park,
Advanced Macroeconomics (G), Seoul National University
Spring 2019, Teaching Assistant to Prof. Soyoung Kim,
Topics in International Monetary Economics (G), Seoul National University
Spring/Fall 2018, Teaching Assistant to Prof. Soyoung Kim,
Macroeconomics (U), Seoul National University

Research Experience:

Research Assistant to Prof. Nicholas Ryan, Yale University, 2026
Research Assistant to Prof. Costas Arkolakis, Yale University, 2021–2022, 2024
Research Assistant to Prof. Lorenzo Caliendo, Yale University, 2023
Research Assistant to Profs. David Atkin and Amit Khandelwal, Yale University, 2022–2023
Research Assistant to Prof. Soyoung Kim, Seoul National University, 2018–2020

Work Experience:

Military Service, Republic of Korea Army (2014–2015)

Working Papers:

"The Spatial Consequences of Industrial Environmental Regulation", (Oct 2026),
Job Market Paper
"Deep Industrial Decarbonization: Theory and Evidence from South Korea"
with Costas Arkolakis and Emmanuel Murray Leclair, (Jan 2026), *Working Paper*
"Public and Private Investment in Electric Vehicle Charging Infrastructure"
with Seung-Yong Yoo, (Oct 2026), *Working Paper*
"An Anatomy of Multinational Production: Evidence from Affiliate Labor Composition"
with Wei Xiang, (Oct 2026), *Working Paper*

Work In Progress:

"Effects of Tariff Shocks under the International Financial Trilemma"
with Yeon Jik Lee, (Aug 2026)
"The Energy Transition of Buildings: Evidence from Korea", (Aug 2026)

Seminar and Conference Presentations:

2026: Southern Economic Association (scheduled),
Society for Economic Dynamics,
LSE Environment Camp.
2025: The 2025 BK21-Jiam Conference on Big Data and Economics,
Urban Economics Association,
Econometric Society World Congress,
Society for Economic Dynamics.

Professional Service:

Referee: *Journal of International Economics, Research Policy*
Organizer: Young Economists Symposium, Yale University, 2022

Languages:

Korean (native), English (fluent), German (beginner)

References:

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Dissertation Abstract

The Spatial Consequences of Industrial Environmental Regulation [Job Market Paper]

How do regulations on local air pollution affect emissions and production when firms can shift pollution across plants, sectors, and places? This question matters because such regulations often target large polluters in densely populated, heavily polluted regions, where pollution damages are

greatest. Yet such targeting creates a problem for policy evaluation: emissions declines among regulated plants may reflect both true abatement and shifts of pollution elsewhere. Comparisons of regulated and unregulated units can therefore overstate emissions reductions. Moreover, because these are local air pollutants rather than greenhouse gases, the aggregate pollution damage can fall with pollution shifts to low-damage places.

I study South Korea's 2020 geographic expansion of its cap-and-trade program for local air pollutants: nitrogen oxides, sulfur oxides, and total suspended particulates. Measuring pollution shifts is challenging because it is often difficult to find suitable control plants due to coverage limitations, and data rarely link emissions, production, and abatement technology. I address these challenges by combining national administrative records and censuses on emissions, sales, fuels, and emission-control equipment from 2016 to 2023, covering all regulated and unregulated manufacturing and power plants in Korea.

Using difference-in-differences, I first show that manufacturing emissions fall by 33–37% across pollutants in regulated districts, relative to the same industries in unregulated districts. Regulated manufacturing plants exhibit larger increases in electricity use, electricity's share of energy use, and emission-control equipment use than matched unregulated unaffiliated plants. The evidence is consistent with pollution shifts to unregulated plants within the same manufacturing firms, to power generation, and to nearby unregulated districts, suggesting that spillovers may affect the comparison units.

To translate these relative effects into aggregate effects when policy exposure and pollution damages vary across places, I develop a spatial model of manufacturing and upstream power plants. The model features the abatement and pollution shift channels above and is calibrated to match the potentially contaminated reduced-form estimates. I find that **g**eographic pollution shifts account for 19% of the manufacturing emissions difference-in-differences estimate. I also use the model to assess aggregate impacts. The actual policy reduces pollution damage by 38% at the cost of a 1% decline in consumption. Regulated districts experience lower pollution damages but larger consumption losses, while unregulated districts experience higher pollution damages and smaller consumption losses. This reduction in pollution damage reflects the concentration of emissions reductions in high-damage locations. Holding national emissions totals fixed for each pollutant, a nationwide alternative with higher emissions prices in districts with greater population exposure achieves larger damage reductions at a lower consumption cost than the actual policy.

Deep Industrial Decarbonization: Theory and Evidence from South Korea,
with Costas Arkolakis and Emmanuel Murray Leclair

Industrial electrification is central to decarbonization, but little is known about which policies induce firms to electrify or when such switches meaningfully reduce emissions. We study these questions in South Korea, where the share of manufacturing plants relying exclusively on electricity rose from 35 to 65% between 2016 and 2022 amid the introduction of environmental regulations and green-technology subsidies. However, emissions reductions from manufacturing remain limited because the largest polluters are least likely to electrify. Using new establishment-level data on fuel use and emissions, we examine whether this electrification was driven by regulations or subsidies and forecast their long-run effects. We develop and estimate a dynamic

spatial model in which firms pay fixed costs to adopt or drop fuels, regulations affect local fuel prices, and subsidies reduce clean-fuel adoption costs. The quantitative results attribute most electrification to subsidies targeted at small and medium-sized firms rather than to environmental regulations. We are extending the model to study how firm growth changes the effects of electrification policies on greenhouse gas emissions in the long run.

Public and Private Investment in Electric Vehicle Charging Infrastructure,
with Seung-Yong Yoo

We study how policies on electric vehicle (EV) charging infrastructure affect charging network deployment and EV adoption in South Korea. We combine vehicle and charger registries, charging transactions, and policy rollouts to evaluate two interventions: subsidies for private charger installation and direct government charger provision. Using instrumental variable approaches, we find that policy-supported chargers crowd in private charger entry. The effect is stronger for slow chargers than for fast chargers. We also show that EV adoption responds most strongly to the availability of slow home chargers and the spatial availability of fast chargers. To account for spatial spillovers, we develop a dynamic spatial model of vehicle and charging markets, which we are currently estimating. We will use the estimated model to evaluate alternative combinations of charger policies.

An Anatomy of Multinational Production: Evidence from Affiliate Labor Composition,
with Wei Xiang

This paper studies how firm productivity is organized. Using Korean foreign affiliate surveys, we show that the executive-manager layer of expatriates (expats) is a more prevalent home-linked connection than intra-firm trade or capital payments. Expats are concentrated in executive and managerial positions, and this upper layer typically persists as local management expands. A labor-hierarchy framework connects these staffing patterns to the organization of firm productivity. Expat-based productivity is higher in more distant and richer host countries and in industries with greater management and administration requirements. We develop a multinational production model with labor hierarchy to quantify how expat organization contributes to cross-country productivity differences.